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The Editors
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Dear Editors,

Please consider our manuscript “Trustworthy prediction intervals after moving an NIR calibration to a new instrument or season: how many reference samples, and from how many batches?” for publication as a Research Paper in *Analytica Chimica Acta*.

Many laboratories move NIR calibrations: to a new handheld unit, to a second production spectrometer, or to a new harvest season. The move is usually checked by the prediction error on a few reference samples, but the results are reported, and products accepted or rejected, with an uncertainty range. We asked whether that range can still be trusted after the move, and what the reference samples have to look like for it to be trusted.

On three public data sets the answer to the first question was no. For uncertainty ranges taken over from the original calibration, the coverage of nominal 90% intervals fell as low as 0.03. The reference samples needed for the transfer are enough to restore a valid range: the transfer is re-fitted with each sample left out in turn, and no extra measurements are needed. Simple Gaussian and regression intervals built from the same left-out errors were nearly as good at 90%, and the paper says so.

The finding we consider most useful for practice concerns where the reference samples come from. With mango data that record the consignment of every fruit, 20 reference fruit from two consignments gave intervals that were too narrow for fruit from other consignments in 6 of 8 instrument and season settings. Gaussian and split-sample intervals failed in the same way. We then confirmed this in 14 instrument–season settings on 11 further handheld units not analysed before, with predictions written down before the runs: the sample-level intervals were too narrow for new consignments in 14 of 14 settings. Spreading the same 20 fruit over many consignments, or leaving out whole consignments when estimating the errors, avoided the problem; with whole consignments left out, the jackknife+ and the classical Gaussian and random-effects formulas all performed well. A simple random-effects model describes how the loss grows as fewer batches are sampled; it was optimistic for two batches, which we report. In a harvest-maturity check of mango at the recommended minimum of 14% dry matter, deciding on the predicted value alone passed 36% of immature fruit; intervals from well-spread reference fruit reduced this to 3.2% (against 9.8% with fruit from two consignments), at the cost of re-measuring about half of the fruit. A fixed guard band around the threshold does about as well; the intervals choose its width from a stated coverage. The rule that follows, take transfer standards from many batches, costs nothing and can be applied

immediately. To make it easy to use, we provide a Python package and a browser tool that plan the sampling and compute the intervals and accept/reject decisions.

The work fits the journal's interest in chemometric methods with important applications. It addresses measurement uncertainty of transferred analytical methods, it is phrased in method-validation terms, and it ends with a concrete procedure. We do not claim new statistical theory: the interval construction is an application of the jackknife+ and of its batch-level version, which we cite. The contribution is the evidence on when such intervals hold after calibration transfer, the comparison with the methods laboratories actually use, and the resulting sampling rule. The batch evidence comes from one product, which we state as a limitation.

All data are public. All code and raw results, and the scripts that regenerate every number, table and figure, will be made publicly available. The manuscript has not been published and is not under consideration elsewhere, and all authors have approved it. The authors declare no competing interests. An AI assistant was used for literature search support, code development, data analysis and drafting, as declared in the manuscript; the authors take full responsibility for the content.

Suggested reviewers: [to be provided by the authors]

Yours sincerely,

[Corresponding author name], on behalf of all authors